Ben Mackert
University of Colorado Boulder
ECEN 3730
Board 3 Report

Summary

Board 3 is my "Golden Arduino": a two layer, 3.9 in by 2.9 in PCB that does the same job as an Arduino Uno (ATmega328P microcontroller, CH340G USB to serial chip, Uno header pin locations) but uses the layout rules from this course to keep switching noise down. It has a solid ground plane on the whole bottom layer, no copper fill on top, decoupling capacitors next to the chips and at the header pins, a ferrite filter on AVCC, isolation jumpers, and labeled test points.

The board was bootloaded and ran the Lab 20 noise sketch with the switching noise shield plugged in. I then made the same shield measurements on my board and on a commercial Whadda Uno. The biggest difference is radiated noise: a pickup loop under my board saw about 12 mV, compared with 173 mV under the commercial board, so my board puts out about 7% as much near field noise. Inside the chip, the quiet HIGH sagged 0.10 V instead of 0.30 V (33%), but the quiet LOW ground bounce was only a little better (73% to 79% of the commercial board). When the shield's slammer pulled 345 mA from the 5 V rail, the dip on the rail was about the same on both boards (0.6 V), but my board overshot more when the current shut off (0.83 V versus 0.2 V). I explain why in the Analysis.

Project Overview and Plan of Record

Plan of Record (POR)

The goal was a board that works as an Uno with any shield, and that is quieter and easier to test than the commercial Uno. The board contains:

1. **Power.** Mini USB (J1) or a 5 V barrel jack (P1), picked by a 3 pin jumper (SLCT) so both can never connect at once; 22 µF bulk (C1) and LED1. An AMS1117-3.3 LDO (PS1, C12, LED2) feeds only the 3V3 header pin, as on the Uno.
2. **Isolation jumpers.** "+5V CH430" and "+5V 328" power each chip separately for debug, with decoupling on the chip side.
3. **USB to serial.** CH340G (U2), 12 MHz crystal, SRV05-4 TVS (D1) on the data lines, TX and RX LEDs.
4. **Microcontroller.** ATmega328P (U1), 16 MHz crystal, 22 µF at each VCC pin (C5, C6), AVCC through a 10 µH ferrite (L1) with 22 µF (C7), 1 µF on AREF (C16).
5. **Reset and programming.** Button with 10 kΩ pull up and 47 nF debounce, DTR auto reset, and a 6 pin ICSP header (J8).
6. **Headers.** Uno sockets in the standard locations, a second row 300 mil outside them tied to ground, 22 µF (C2, C13) at the 5V and 3V3 pins, and LED5 on D13.
7. **Test points.** 4 hole test points for a 10x probe with a ground spring: +5V, +3V3, TP D13 (trigger) and RESET.

What I changed from the commercial Uno

Before starting I looked at the commercial Whadda Uno for things that make noise, and planned three changes:

1. **One solid ground plane, no top pour.** The commercial board has copper pour on both layers, and the top pour is chopped into small islands by traces. My bottom layer is one unbroken ground plane, so every signal's return current runs right underneath it.
2. **Capacitors at the pins.** The commercial board has no capacitor right at each VCC pin. Mine has 22 µF at both VCC pins and a separate filtered AVCC, so switching current comes from right next to the chip instead of through the board's power traces.
3. **A ground pin next to every header signal.** The commercial digital header has one ground pin. My extra row of ground pins gives a shield's signals a return path right beside them.

Key parts

Ref	Part	Datasheet
U1	ATmega328P, TQFP-32	https://www.microchip.com/en-us/product/ATmega328P
U2	CH340G USB to UART	https://www.wch-ic.com/downloads/CH340DS1_PDF.html
PS1	AMS1117-3.3 LDO	https://www.lcsc.com/product-detail/C6186.html
D1	SRV05-4 TVS array	https://www.onsemi.com/pdf/datasheet/srv05-4-d.pdf

Table 1: Main parts. All passives, LEDs and the ferrite are 1206.

Design calculations and power budget

AVCC filter. L1 (10 µH) and C7 (22 µF) make a low pass filter with a corner at $1 / (2\pi\sqrt{LC})$ = 11 kHz, so switching noise from the digital supply is mostly blocked from the ADC supply. AVCC draws well under 1 mA, far below L1's 25 mA rating.

Reset and LEDs. 10 kΩ × 47 nF = 0.47 ms ignores switch bounce. Each LED has 1 kΩ: (5 V − 2 V) / 1 kΩ = 3 mA.

Shield loads (what the board must drive). D12, D11 and D10 each drive an LED and a 49 Ω resistor on the shield. With a 2 V LED and about 25 Ω inside the output pin, I expected (5 − 2) / 74 Ω = about 40 mA per pin, or 120 mA for three pins switching together. The slammer pulls 5 V / 10 Ω = up to 500 mA from the 5 V rail through a MOSFET.

Power budget. The chips and LEDs use about 30 mA. The shield adds 120 mA for a few microseconds and the slammer about 350 mA for 400 µs every 11.4 ms (12 mA average), so the average is under 50 mA and USB power is fine.

What it means for this board to work

S1. +5V reads about 5 V from either USB or the barrel jack (picked by SLCT) and LED1 is on.
S2. The 3.3 V LDO is on (LED2 lit).
S3. The computer sees the CH340G as a COM port, and the bootloader burns through the ICSP header.
S4. Blink uploads over USB and can be changed to flash D13 at 5 Hz.
S5. With the noise shield, D13 triggers the scope and D12 to D10 each drive about 40 mA.
S6. Quiet LOW and quiet HIGH noise are 20% to 50% of the commercial board's (lab manual goal).
S7. The 5 V rail noise with the slammer is lower than on the commercial board.
S8. Near field emissions are much less than 10% of the commercial board's (lab manual goal).

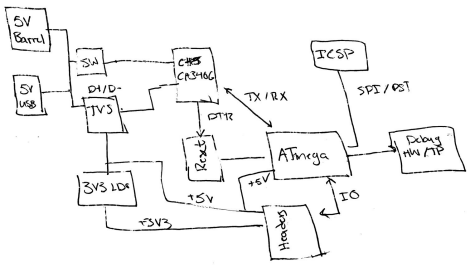


Figure 1: Hand sketch of the POR block diagram.

Test plan, risks and schedule

Test	Setup and probe point	Pass if	Spec
T1	Both jumpers out; +5V and +3V3 TPs, USB then barrel jack	4.75 to 5.25 V, LED1 and LED2 on	S1, S2
T2	+5V CH430 in; USB to laptop	COM port appears, TX/RX LEDs flash	S3
T3	+5V 328 in; Uno as ISP on J8, then USB upload; TP D13	Bootloader burns, Blink runs at 5 Hz	S3, S4
T4	Shield and Lab 20 sketch; D13 on CH1, shield pins on CH2	LED current, quiet LOW, quiet HIGH and board 5 V recorded	S5, S6
T5	Same; slammer 10 Ω on CH2, 5 V rail or quiet HIGH on CH1	Rail dip and overshoot recorded on both edges	S7
T6	Same; 10x probe tip shorted to its ground as a loop under the board	Peak pickup recorded on both boards	S8

Table 2: Test plan. T4 to T6 are repeated on the commercial Uno with the same setup.

Risks. Wrong footprints: all 1206, headers checked against the Uno drawing. A short or bad chip: isolation jumpers. A bootloader that will not burn: the ICSP header. ESD on USB: the TVS. Fab rejection: DRC and JLCPCB DFM.

Step	Date	What I did
1. Plan of record	2026-10-13	Features, specs S1 to S8, test plan and the three changes above
2. Parts and BOM	2026-10-13 to 10-20	Picked the 328, CH340G, crystals, TVS and LDO; all passives 1206
3. Schematic	2026-10-13 to 10-19	Altium schematic in blocks (figure 2)
4. Layout	2026-10-18 to 10-20	Layout, DRC, Gerbers; JLCPCB order Y5-13373447A approved 10-20
5. Assembly	2026-11-02	Boards arrived and were hand soldered the same day
6. Bring up and test	2026-11-02 to 11-04	Worked the first time, no hard errors or rework; noise tests 11-04
7. Documentation	2026-11-10	This report

Table 3: Schedule, following the seven board design steps.

Schematic and layout

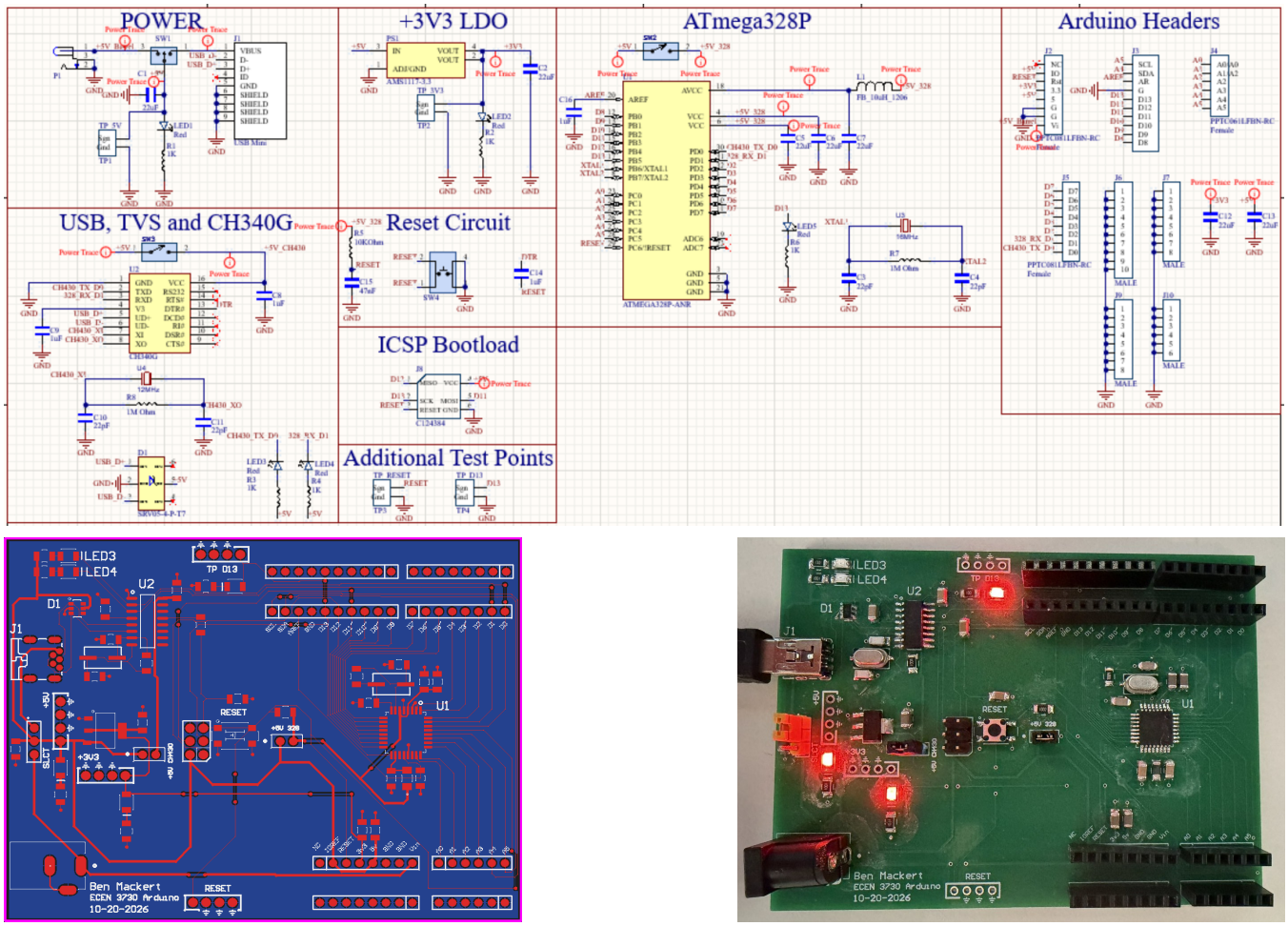


Figure 2: Altium schematic (top), final layout (bottom left; top copper red, bottom ground pour blue) and the assembled board in the same orientation, running from USB with LED1, LED2 and the D13 LED on (bottom right).

- **Ground.** The bottom layer is one solid ground pour and the top has no copper fill. Every ground pin goes straight down to the plane with a via. A few short signal jumpers had to go on the bottom layer; the longest cuts the plane for about 8 mm.
- **Decoupling.** C5 and C6 sit just above U1, C12 is at the LDO, and C2 and C13 are right at the 5V and 3V3 header pins.
- **Traces.** 20 mil for power, 6 to 10 mil for signals; crystals next to the pins they drive.
- **Labels.** Every pin, jumper and test point, plus my name and the date.
- **DRC.** 0 clearance, short or unrouted errors; only the barrel jack slot size and one mask sliver were flagged, and JLCPCB accepted the board.

Bring Up and Measurements

All measurements used 10x probes with the spring ground tip, DC coupled, with the bandwidth limit off. For T4 to T6 the noise shield was plugged into the board under test, running the Lab 20 sketch: D13, D12, D11 and D10 turn on together for a short pulse while D9 is held low (quiet LOW) and D8 is held high (quiet HIGH); 1 ms later D7 turns on the slammer for 400 µs, and the loop repeats every 11.4 ms.

Power, CH340G and bootload (T1 to T3; S1 to S4)

With USB power, LED1 (+5V), LED2 (+3V3) and the D13 LED all came on (figure 2), and the scope showed the board rail at 5.12 to 5.16 V while the board was running. The board ran the Lab 20 sketch uploaded over USB, so the CH340G, the bootloader and the upload path all worked. With SLCT moved to the barrel jack, the 5 V adapter powered the board as well. Blink, changed to flash the D13 LED at 5 Hz, uploaded and ran. I did not record a voltage on +3V3; LED2 lit, which shows the LDO output was up. The board worked the first time it was powered: there were no hard errors and no rework.

Three outputs switching (T4; S5, S6)

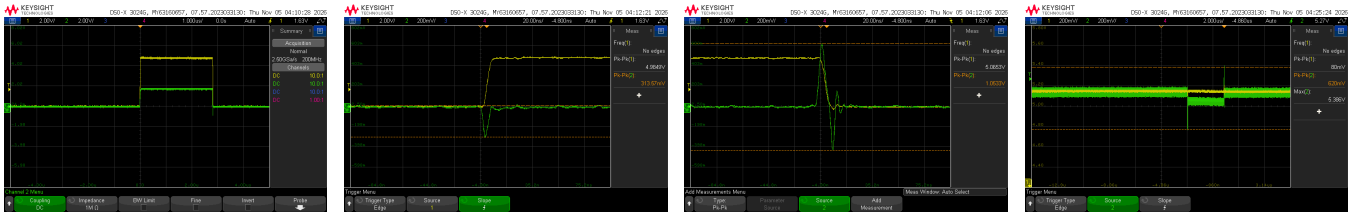


Figure 3: My board, triggered on D13. Left to right: (1) D13 yellow and the voltage across one 49 Ω shield resistor green, 1 µs/div; (2) quiet LOW green at 200 mV/div, rising edge, 20 ns/div; (3) quiet LOW, falling edge; (4) board 5 V rail yellow and quiet HIGH green, both 200 mV/div, 2 µs/div.

LED current. The voltage across a 49 Ω resistor was 1.72 V (capture 1), so each pin drives 1.72 V / 49 Ω = 35 mA and the three pins together switch about 105 mA. That is a little under my 40 mA estimate. The pulse lasts 2.8 µs.

Quiet LOW (ground bounce). D9 should stay at 0 V. When the outputs turn on it dipped to −0.30 V (0.31 V peak to peak). When they turn off it jumped to +0.62 V and rang down to −0.43 V (1.05 V peak to peak), and settled in about 30 ns. The turn off edge is noisier because the outputs switch off faster than they switch on (7 ns versus 19 ns on the slammer edges below).

Edge speed. Noise scales with dI/dt, so the edges have to match for a fair comparison. On my board D13 rose in 5.0 ns and fell in 5.6 ns (10% to 90%); on the commercial board it rose in 4.8 ns and fell in 5.1 ns. My edges are about 5% to 10% slower, so a small part of my lower noise comes from the edges, not the layout.

Quiet HIGH versus the board rail. The quiet HIGH pin shows what the 5 V supply looks like inside the chip. It had spikes of −0.38 V and +0.23 V at the edges (0.62 V peak to peak), and sat 0.10 V low for the whole time the 105 mA was flowing. At the same moment the board's 5 V rail moved only 80 mV peak to peak, so almost all of the noise is inside the chip's package.

Commercial board. With the same shield, sketch and scales, the commercial board's quiet LOW dipped 0.40 V on the turn on edge and rang 1.43 V peak to peak on the turn off edge, so mine was 79% and 73% of it. Its quiet HIGH sagged about 0.30 V during the pulse (mine 0.10 V, 33%), and its board 5 V rail sagged 0.10 V along with it, where mine did not visibly move (figure 5, top row).

Slammer on the 5 V rail (T5; S7)

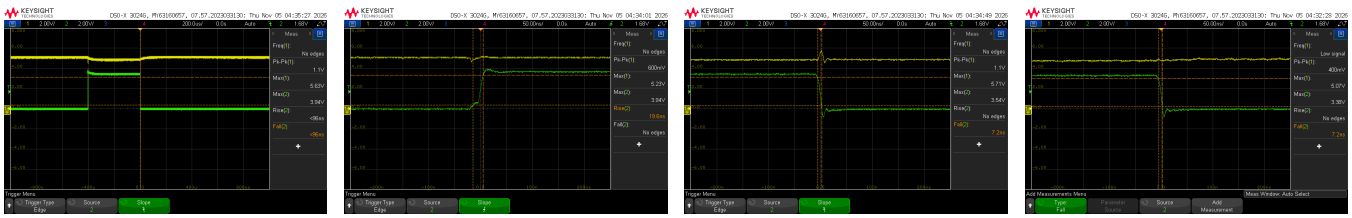


Figure 4: Slammer current (green, voltage across 10 Ω, 2 V/div). Left to right: (1) my board, full 400 µs pulse, 5 V rail yellow; (2) my board, quiet HIGH yellow as the current turns on, 50 ns/div; (3) my board, 5 V rail as the current turns off; (4) commercial Uno, same as (3).

The slammer drew 3.45 V / 10 Ω = 345 mA on my board (with a 3.9 V peak right at turn on) and 340 mA on the commercial board. On both boards the current rose in about 19 ns and fell in about 7.2 ns. Because the current and the edge speeds match, the two boards can be compared fairly. During the pulse my 5 V rail sat 0.24 V lower than before, so the supply behind it looks like about 0.24 V / 0.345 A = 0.7 Ω.

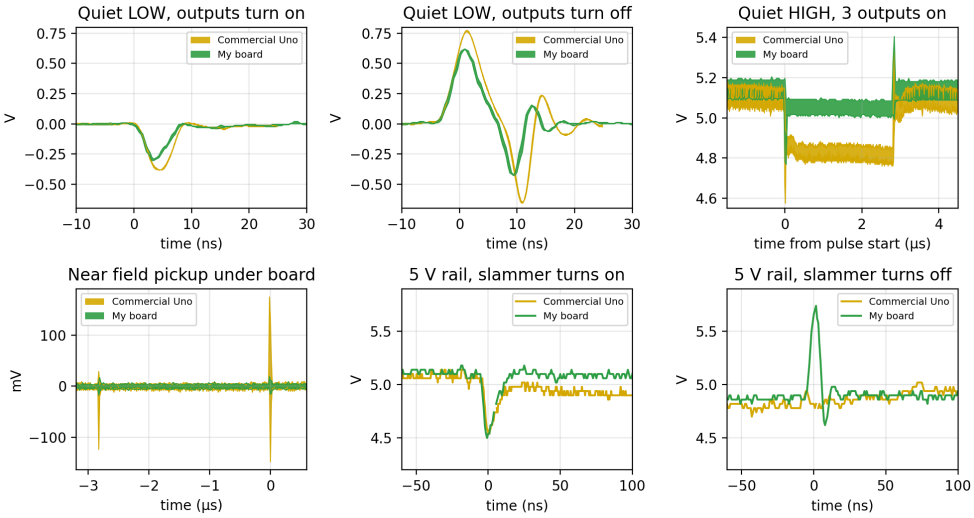


Figure 5: My board (green) and the commercial Uno (yellow), read off the screenshots and lined up on the switching edge. Top: quiet LOW on both edges and quiet HIGH during the 2.8 µs pulse. Bottom: near field pickup under each board, and the shield 5 V rail when the slammer turns on and off.

Near field emissions (T6; S8)

Figure 5 (bottom left) shows the pickup loop at the worst spot under each board, at 50 mV/div, with the setup in figure 6. Away from either board it read 6 mV, the scope's own noise floor. Under the commercial board it picked up spikes of +173 mV and −148 mV; under mine the largest was about 12 mV, barely above the floor and about 7% of the commercial board. Table 4 collects every noise result.

Measurement	My board	Commercial	Mine / Comm.
D13 rise / fall time (10–90%)	5.0 / 5.6 ns	4.8 / 5.1 ns	1.0 to 1.1
Quiet LOW, on edge (pk-pk)	0.31 V	0.40 V	0.79
Quiet LOW, off edge (pk-pk)	1.05 V	1.43 V	0.73
Quiet HIGH sag during pulse	0.10 V	0.30 V	0.33
Quiet HIGH, 3 pins (pk-pk)	0.62 V	0.84 V	0.74
Board 5 V rail sag during pulse	none seen	0.10 V	–
Slammer current, rise / fall time	345 mA, 19 / 7.4 ns	340 mA, 19 / 7.2 ns	1.0
5 V dip, slammer on	0.64 V	0.55 to 0.62 V	1.1
5 V overshoot, slammer off	0.83 V	0.22 V	3.8
Quiet HIGH, slammer on / off	−0.48 / +0.36 V	not captured	–
Near field peak pickup	12 mV	173 mV	0.07

Table 4: Noise results, same shield, sketch, probes and scales on both boards.

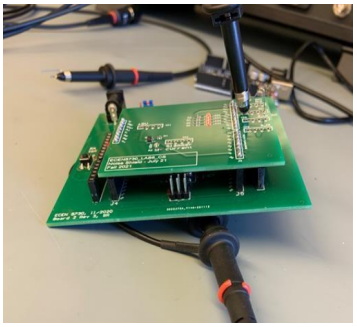


Figure 6: Near field setup: probe tip shorted to its ground spring as a loop under the board.

Specification scorecard

Spec	Test	Expected	Measured
S1: +5V, USB or jack	T1	about 5 V, LED1 on	5.12 to 5.16 V; jack works
S2: 3.3 V LDO on	T1	LED2 on	yes
S3: COM port and bootloader	T2, T3	yes	yes (runs sketches)
S4: Blink at 5 Hz	T3	yes	yes
S5: shield pin current	T4	about 40 mA	35 mA per pin
S6: quiet pins vs commercial	T4	20 to 50%	HIGH 33%; LOW 73 to 79%
S7: 5 V rail vs commercial	T5	lower	same dip, larger overshoot
S8: near field vs commercial	T6	much less than 10%	7%

Table 5: Specification versus measurement.

Analysis

Why the noise is what it is

All of this noise comes from the same rule as Board 2: an inductance fights a change in current by making a voltage, V = L × dI/dt, so a fast change through even a few nanohenries makes a noticeable voltage.

Near field. A current flowing around a loop makes a magnetic field, and a bigger loop leaks more of it. On my board every signal runs over the solid plane, so its return current flows right underneath it and the loop is tiny. On the commercial board the returns detour around a top pour broken into pieces by traces, so the loops are bigger and the same 105 mA edges radiate about 14 times more. This is the clearest win for the layout.

Quiet HIGH. The 105 mA has to come from somewhere. On the commercial board part of it came through the board's power path, so both the board rail and the chip sagged. On mine the 22 µF capacitors at the VCC pins supplied it, so the board rail did not move and the chip sagged a third as much.

Quiet LOW. The ground bounce improved only to 73% to 79% because it is set mostly by the inductance of the chip's own ground pins and bond wires. That is the same part on both boards; my plane only shortened the part of the path that is on the board.

Slammer. At turn on, dI/dt = 345 mA / 19 ns = 18 mA/ns, and a 0.64 V dip means about 0.64 V / 18 mA/ns = 35 nH between the slammer and the nearest capacitor. Most of that path is on the shield and the header pins, the same for both boards, so the dips matched. At turn off the current stops 2.5 times faster and my rail jumped 0.83 V and rang, against 0.2 V on the commercial board. My best explanation is that my board has only ceramic capacitors, which have almost no internal resistance to soak up the energy, while the commercial board most likely has electrolytic capacitors that damp it.

What worked. The solid plane with no top fill (7% near field), the capacitors at the pins (33% quiet HIGH sag), the isolation jumpers and ICSP header for bring up, and using the same shield, sketch, probes, scales and matched edge speeds on both boards so the comparison is fair.

What I would do differently

- **Damp the 5 V rail.** Add a bulk capacitor with some resistance (electrolytic or tantalum, or a small resistor in series with a large ceramic) at the 5V header pin, so the rail does not ring when a shield load shuts off.
- **Measurement scale.** The slammer captures were taken at 2 V/div, where one pixel is 40 mV (the same mistake as Board 2). Next time I would use 200 mV/div with a 5 V offset.
- **Layout and test.** Route the few bottom jumpers on top so nothing cuts the plane, and add a current sense resistor and test points on TX/RX and USB D+/D−.

Takeaways. A solid ground plane under every signal cut radiated noise 14 times, and decoupling right at the pins cut the chip's supply sag to a third. Neither can fix noise inside the package, and a rail feeding a load that can switch off suddenly needs damping, not just capacitance.